Rental Price Prediction Using Multiple Linear Regression

Product Backlog

A **Product Backlog** is a prioritized list of all features, functionalities, enhancements, and technical tasks required to successfully develop the Rental Price Prediction System. It serves as the primary planning document in Agile development and is continuously updated throughout the project lifecycle.

Backlog ID	User Story / Requirement	Description	Priority	Estimated Sprint
PB-01	Project Planning	Define project objectives, scope, timeline, and required resources.	High	Sprint 1
PB-02	Requirement Gathering	Identify functional and non-functional requirements for the system.	High	Sprint 1
PB-03	Dataset Collection	Collect historical rental property data including area, bedrooms, bathrooms, age, and rent.	High	Sprint 1
PB-04	Data Cleaning	Remove missing values, duplicate records, and inconsistent entries from the dataset.	High	Sprint 1
PB-05	Data Preprocessing	Convert categorical values, normalize data if necessary, and prepare the dataset for training.	High	Sprint 1
PB-06	Feature Selection	Select important independent variables affecting rental price.	High	Sprint 2
PB-07	Exploratory Data Analysis	Analyze relationships between variables using graphs and statistical methods.	Medium	Sprint 2
PB-08	Split Dataset	Divide data into training and testing datasets.	High	Sprint 2
PB-09	Develop Regression Model	Implement Multiple Linear Regression using Scikit-learn.	High	Sprint 2

Backlog ID	User Story / Requirement	Description	Priority	Estimated Sprint
PB-10	Train Model	Train the regression model using historical rental data.	High	Sprint 2
PB-11	Model Testing	Test the model using unseen test data.	High	Sprint 3
PB-12	Performance Evaluation	Calculate MAE, MSE, RMSE, and R^2 Score to measure prediction accuracy.	High	Sprint 3
PB-13	User Interface Design	Design an attractive and user-friendly interface using Tkinter or Flask.	Medium	Sprint 3
PB-14	Input Validation	Validate user inputs such as area, bedrooms, bathrooms, and age.	Medium	Sprint 3
PB-15	Rental Price Prediction	Predict rental price based on user-entered property details.	High	Sprint 3
PB-16	Display Prediction Result	Show predicted rental price clearly on the screen.	High	Sprint 3
PB-17	Error Handling	Handle invalid inputs and unexpected system errors gracefully.	Medium	Sprint 4
PB-18	Data Visualization	Display graphs such as feature correlation and regression plots.	Medium	Sprint 4
PB-19	Model Optimization	Improve prediction accuracy through parameter tuning and feature engineering.	Medium	Sprint 4
PB-20	System Testing	Perform unit testing, integration testing, and user acceptance testing.	High	Sprint 4
PB-21	Documentation	Prepare project report, user manual, and technical documentation.	High	Sprint 4
PB-22	Final Deployment	Deploy the application for end users.	Medium	Sprint 4

Backlog ID	User Story / Requirement	Description	Priority	Estimated Sprint
PB-23	User Feedback	Collect user feedback for future improvements.	Low	Sprint 5
PB-24	Future Enhancement	Add additional features such as location, parking, furnishing, and nearby amenities for improved prediction accuracy.	Low	Sprint 5